

Figuring out possible outcomes

A table can be used to work out all the possible outcomes of two combined events. For example, if two dice are rolled, their scores will have a combined total of between 2 and 12. There are 36 possible outcomes, which are shown in the table below. Read down from each red dice and across from each blue dice for each of their combined results.



		red dice throws						
blue dice throws	Red							
	Blue		2	3	4	5	6	7
		3	4	5	6	7	8	
		4	5	6	7	8	9	
		5	6	7	8	9	10	
		6	7	8	9	10	11	
		7	8	9	10	11	12	

6 ways out of 36 to throw 7, for example blue dice rolling 1 and red dice rolling 6

5 ways out of 36 to throw 8, for example blue dice rolling 2 and red dice rolling 6

4 ways out of 36 to throw 9, for example blue dice rolling 3 and red dice rolling 6

3 ways out of 36 to throw 10, for example blue dice rolling 4 and red dice rolling 6

2 ways out of 36 to throw 11, for example blue dice rolling 5 and red dice rolling 6

1 way out of 36 to throw 12

KEY

Least likely
The least likely outcome of throwing 2 dice is either 2 (each dice is 1) or 12 (each is 6). There is a $\frac{1}{36}$ chance of either result.

Most likely
The most likely outcome of throwing 2 dice is a 7. With 6 ways to throw a 7, there is a $\frac{6}{36}$, or $\frac{1}{6}$, chance of this result.